

Effect of a Unique Dwarfism on Growth, Production, and Reproduction Performance of the Nicobari Chicken Breed

Santosh Haunshi, Shanmugam Murugesan, Prince L.L. Leslie, Rajkumar Ullengala and Chatterjee R. Nath

ICAR-Directorate of Poultry Research, Rajendranagar, Hyderabad-500030, Telangana, India

The present study aimed to determine the inheritance and effect of dwarfism, which occurs naturally in the Nicobari chicken breed, on various growth and production traits. The dwarf character did not affect body weight for up to 4 weeks; but had a negative impact from 8 weeks onward, with body weight declining by 5.5%–9.0% in males and 9.0%–16% in females. Shank length was 22% and 27% shorter in male and female dwarf birds, respectively, than in full-size birds. The average age at first egg was 3.08% higher in dwarf hens than in their normal counterparts, and was accompanied by a non-significant decline (5.63%) in 40 weeks egg production. Egg weights were 2.23% and 2.64% lower at 32 and 40 weeks, respectively, in dwarf hens. Instead, the shape index and shell thickness were better in dwarf hens. The eggs of dwarf hens were darker and red, with less hue than those of normal hens; whereas internal egg quality was the same. Fertility and hatchability were lower in dwarf dams mated with dwarf sires than in crosses or reciprocal crosses between normal and dwarf birds. Breeding experiments and sequencing of transmembrane protein 263 ruled out the sex-linked (*dw*) and autosomal dwarfism (*adw*), as the genetic causes of dwarfism in the Nicobari breed, which instead appears to follow a novel pattern and has a unique effect on growth traits.

Key words: dwarfism, egg quality, Nicobari breed, production, reproduction,

J. Poult. Sci., 62: jpsa.2025012, 2025

Introduction

Indigenous chicken breeds in India present unique characteristics, such as a naked neck, dwarfism, frizzle, and fibromelanosis. These characteristics favor survival and performance in a tropical environment, and the underlying genes responsible for these characters have been already identified[1–3]. While the effect of naked neck character on production, physiological, and immunocompetence traits has been extensively investigated; the dwarf character of the Nicobari breed remains poorly studied. Even the autosomal or sex-linked inheritance of dwarfism in this breed remains unclear. The three most common types of dwarfism in chickens are sex-linked (*dw*), autosomal (*adw*), and bantam[4], in addition to the creeper (*Cp*) trait[5]. Sex-linked dwarf-

ism is the most extensively studied form of dwarfism in chickens. It originates from deletions or mutations in the growth hormone receptor gene and affects various economically important traits, such as body weight, shank length, and egg production[6–8]. Similarly, autosomal dwarfism results from a mutation in the transmembrane protein 263 (TMEM263) gene negatively affects body weight, shank length, and production traits[9,10].

Nicobari, a registered native chicken breed, originates from the Nicobar Islands off the coast of India, where this slow-growing indigenous breed was developed by local tribes[11]. Nicobari birds have compact bodies, short legs, single combs, red wattles, white, red or a mixture of white and red ear lobes, white skin, white shanks, and orange-red eyes. Both male and female birds show normal feather morphology and feather distribution, as well as rapid (fast) feathering. Female birds have matte brown colored plumage; whereas males display reddish-brown plumage with golden yellow hackles and saddles, along with shiny bluish-black colored sickle feathers. Dwarf Nicobari birds have shorter shanks and smaller body size compared to non-dwarf Nicobari birds. Their relatively smaller size implies body weights of 1.3 to 1.4 kg at 40 weeks of age. Hen-housed egg production ranges from 159 to 169 eggs up to 72 weeks[12,13]. Conservation, improvement, and use of native chicken breeds is given increasing

Received: October 29, 2024, Accepted: February 5, 2025

Available online: March 10, 2025

Correspondence: Santosh Haunshi, ICAR-Directorate of Poultry Research, Rajendranagar, Hyderabad-500030, Telangana, India (Email: Santosh.Haunshi@icar.gov.in)

The Journal of Poultry Science is an Open Access journal distributed under the Creative Commons Attribution-NonCommercial-Share-Alike 4.0 International License. To view the details of this license, please visit (<https://creativecommons.org/licenses/by-nc-sa/4.0/>).

Table 1. Growth traits of dwarf and normal male birds.

Traits	N	Dwarf	N	Normal	P value	Change (%)
Body weight (g)						
0 day	187	31.84 ± 0.22	380	31.71 ± 0.14	0.447	-0.41
4 weeks	185	207.9 ± 3.41	376	208.3 ± 2.14	0.563	0.19
8 weeks	96	547.3 ± 15.8 ^b	348	577.0 ± 6.09 ^a	0.0001	5.43
12 weeks	139	943 ± 14.8 ^b	299	1018 ± 8.62 ^a	0.0001	7.95
16 weeks	93	1213 ± 17.9 ^b	295	1334 ± 9.92 ^a	0.0001	9.07
20 weeks	168	1434 ± 16.0 ^b	170	1563 ± 13.9 ^a	0.0001	8.25
40 weeks	103	2096 ± 26.2 ^b	70	2219 ± 26.7 ^a	0.0001	5.54
Shank length (mm)						
8 weeks	72	58.14 ± 1.07 ^b	245	72.83 ± 0.39 ^a	0.0001	25.27
12 weeks	139	73.76 ± 0.56 ^b	299	94.07 ± 0.32 ^a	0.0001	27.54
20 weeks	168	82.34 ± 0.41 ^b	170	108.7 ± 0.36 ^a	0.0001	24.25
40 weeks	103	84.72 ± 0.46 ^b	70	109.6 ± 0.47 ^a	0.0001	22.70

Values are expressed as means ± standard error.

^{a, b} Means with different superscripts within a row differ significantly; N, number of observations.

importance in India. Presently, the breeding populations of the Nicobari chickens are being maintained by the Indian Council of Agricultural Research at Hyderabad, Izatnagar, and Port Blair (Andaman and Nicobar Islands). Around 7000–8000 chickens of this breed are present on the Nicobar Islands[11]. The purpose of breeding the Nicobari breed is to develop this breed as egg type improved native chicken strain for rearing in low-input backyard and free-range systems of production for tribal and rural areas of the country.

At present, the effect of dwarfism on various commercially important traits in this breed is not known. Therefore, this study sought to investigate the effect of dwarfism on growth, production, reproduction, and egg quality in Nicobari chickens, as well as determine whether the trait is inherited in a sex-linked or autosomal manner.

Materials and methods

Experimental design

The experiment was conducted on two generations of Nicobari breed chickens. Growth and production traits were evaluated in the first generation, comprising a pedigree dwarf and normal population, and then in the second generation, progenies from three matings (dwarf × dwarf, dwarf × normal, and normal × normal) of first generation were evaluated for growth and production traits. A total of 1136 chicks from two generations (534 and 602) were housed in four hatches (two per generation). Of these, 569 were female and 567 were male. Day-old chicks were wing-banded and vaccinated against *Marek's disease* at hatchery. Body weight was measured on day 1, and then again for all surviving chicks on the 4th week. Details regarding the number of chicks evaluated for different growth traits are presented in Tables 1 and 2. The discrepancy in sample size between time points was due to assessment of a different number of hatches, routine mortality, and space constraints that reduced the number

of male and female birds housed in cages. Straight-run chicks were reared on the floor (deep-litter system) for up to 15 weeks. Subsequently, separate sex rearing was performed on the floor for 20 weeks. At about 14 to 16 weeks of age, male and female Nicobari birds were separated. Subsequently, the dwarf and normal birds of each sex were identified by observing birds' shank length (height) and body weight (size). Birds that were short and small were identified as dwarf (Figure 1). Data recorded on the shank length and body weight of male and female birds up to 40 weeks of age were used to cross-check and confirm the visual identification of dwarf birds. A pair of normal and dwarf adult cocks are shown in Figure 2.

Males and females were housed in individual cages to record their growth and production traits for up to 40 weeks. The birds were reared in open-sided sheds during the brooding, growing, and laying stages. Chicks were provided with a layer chick starter ration up to 8 weeks of age, a grower ration from 9 to 20 weeks of age, and a layer ration from 21 weeks onward. The rations provided either 2600 kcal/kg metabolizable energy (starter and layer) or 2500 kcal/kg metabolizable energy (grower) and either 18% crude protein (starter) or 16% crude protein (grower and layer). *Ad libitum* feed and round-the-clock access to clean drinking water were available. Light was provided to the hens for 16 h/day. Standard healthcare practices were followed throughout the study. This study was approved by the Institutional Animal Ethics Committee of the ICAR-Directorate of Poultry Research, Hyderabad (IAEC/DPR/20/7).

Growth and production traits

Body weight was recorded to the nearest 0.1 g accuracy on day 0, and then at 4, 8, 12, 16, 20, and 40 weeks of age using a digital balance. Shank length was measured to the nearest 0.1 mm accuracy at 8, 12, 20, and 40 weeks of age using digital Vernier calipers (Mitutoyo). Age at first egg production, egg production up to the age of 40 weeks, and egg weight at 32 and 40 weeks



Fig. 1. (A) Pair of normal cockerel and pullet, and (B) pair of dwarf cockerel and pullet of the Nicobari breed. Dwarf birds have shorter shanks and smaller body size.

were recorded.

Fertility and hatchability traits

The effects of dwarfism on reproductive traits, such as fertility and hatchability, were studied over two generations. Matings were performed following a 1:3 ratio in a pedigree setup. In the first generation, 14 dwarf sires were mated with 42 dwarf dams, 20 normal sires were mated with 60 normal dams, and 25 dwarf sires were mated with 75 normal dams (cross). In the subsequent generation, 22 dwarf sires were mated with 66 dwarf dams, 17 normal sires were mated with 58 normal dams, 20 normal sires were mated with 60 dwarf dams (cross), and 20 dwarf sires were mated with 61 normal dams (reciprocal cross). At the time of regeneration, the ages of the sires and dams ranged from 46 (first generation) to 48 weeks (second generation). Hatched eggs from each dam were collected over 10–12 days. Hatching eggs were fumigated on each day of collection and stored in a cold room (14 to 15 °C). On the day of setting, the eggs were removed from the cold room and kept at room temperature until sweating was over. The eggs from each sire and dam were recorded and placed in the setter. On the 18th day, hatching eggs were removed from the setter and candling was performed to identify and record fertile eggs from each dam. The eggs were placed in pedigree boxes dam-wise and transferred to hatchers. On the 21st day, the chicks were removed from the hatcher and the number of chicks hatched from each dam was recorded. Dam-wise fertility, measured as (number of fertile eggs/total number of eggs) $\times 100$, and hatchability (FES = (number of chicks hatched/number of fertile eggs set) $\times 100$ and TES = (number of chicks hatched/number of total eggs set) $\times 100$, were recorded. Purebreds, crosses, and reciprocal crosses were compared for fertility and hatchability using pooled data from both generations. After collecting the fertility and hatch-

ability data, the breeding experiment in the second generation was discontinued owing to heavy mortality in the flock at 5–8 weeks of age.

Semen quality traits

Semen quality was evaluated by estimating semen volume, motility, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) dye reduction, sperm concentration, and live and abnormal sperm counts, as previously reported[14–18]. Following the massage method, semen from both dwarf and normal cocks housed in individual cages was collected in sterile glass funnels[14]. The semen ejaculate volume was assessed by drawing the sample into a 1-mL syringe with an accuracy of 0.02 mL. Individual samples were diluted four times using a high-temperature diluent[15]. Sperm motility was assessed visually under a light microscope as the percentage of progressively motile spermatozoa. Sperm concentration was estimated using a spectrophotometric method[16]. The MTT dye reduction assay was performed in duplicate to assess the ability for fertilization. MTT reduction was measured at 570 nm using a colorimeter[17]. Live and abnormal sperm were estimated by differential eosin-nigrosin staining[18]. In each slide, a minimum of 200 sperm cells were examined under an oil immersion ($\times 100$) microscope. Both fully and partially pink-stained cells were counted as dead. The same slides were used to estimate the percentage of abnormal sperm cells.

Egg quality traits

Egg quality traits were recorded at 40 weeks of age. External traits included egg weight, shape index, and shell thickness. The length and breadth of the eggs were measured using digital Vernier calipers to calculate the shape index (width/length). Internal traits included yolk index, albumen index, Haugh unit, and yolk



Fig. 2. Pair of dwarf (left) and normal (right) cocks of the Nicobari breed. Note that the dwarf cock has shorter shanks and smaller body size.

color. Yolk height and width were measured to determine the yolk index; whereas albumen height and width were measured to calculate the albumen index. Yolk and shell weights were recorded, and albumen weight was determined by subtracting the yolk and shell weights from the egg weight. Haugh units were calculated using a standard formula. Yolk color was measured using a Roche color fan. The percentages of yolk, albumen, and shell weight, as well as the yolk-to-albumen ratio were calculated.

Eggshell color characteristics were measured at three locations (broad end, sharp end, and equatorial region) on each egg using a color reader (CR-10 Plus; Konica Minolta). Recordings followed the International Commission on Illumination parameters: lightness (L^*), redness (a^*), yellowness (b^*), chroma [$C^* = \sqrt{(a^{*2}) + (b^{*2})}$], and hue [$H^* = \arctangent(b^*/a^*)$]. Shell color index (SCI) was calculated as $SCI = (L^* - a^* - b^*)$ [19]. The eggshell color value (E) was calculated as $E = (L^{*2} + a^{*2} + b^{*2})^{1/2}$. Using the average E values measured at the broad end, sharp end, and equatorial region, the E value of the entire shell color was determined as follows: whole shell color E value = $(E_{\text{Sharp end}} + E_{\text{Equator}} + E_{\text{Broad end}})/3$ [20]. Lower SCI and E values indicated a darker shell color.

PCR and nucleotide sequencing

To determine whether the genetic cause of dwarfism in the Nicobari breed was due to a mutation (*adw*) leading to loss of function in the TMEM263 protein, a 264-bp fragment corre-

sponding to the TMEM263 gene was amplified by PCR from genomic DNA extracted from the whole blood of five dwarf and five normal Nicobari birds. The fragment was sequenced as described by Wu et al. [10] using the following primers: 5'-AGG TTC AAT CAA AGA CCA CCC G-3' (forward) and 5'-CCC GTT AAA GGC ACT TTG CT-3' (reverse). The PCR products were resolved on a 3% agarose gel and their size was determined using 50-bp markers. PCR products were purified and directly sequenced using the Sanger method by Bioserve Biotechnologies Pvt. Ltd., Hyderabad (India). The sequences were deposited to the NCBI GenBank under accession numbers OQ134946, OQ134947, OQ134948, OQ134949, OQ134950, OQ134951, OQ134952, OQ134953, OQ134954, and OQ134955. To determine whether the dwarfism observed in this breed was due to sex-linked deletion in the growth hormone receptor gene (*dw*), a 253-bp fragment corresponding to the receptor gene (deletion allele) was amplified with the following primers: 5'-TTG TTC ACT CTC CAC AAG GC-3' (forward) and 5'-GAT TCT CCT GGC AGA ATC TC-3' (reverse) [21]. The 463-bp normal allele fragment of the receptor was amplified with the following primers: 5'-CTT TGT GAC CAA GGG GAT CAG G-3' (forward) and 5'-GAA TGC TCG ATT CTC CTG GCA G-3' (reverse).

Statistical analysis

The mean and standard error for different traits was calculated

Table 2. Growth and production traits of dwarf and normal female birds.

Traits	N	Dwarf	N	Normal	P value	Change (%)
Body weight (g)						
0 day	221	31.58 ± 0.21	348	31.43 ± 0.14	0.447	-0.47
4 weeks	221	191.4 ± 3.25	344	194.3 ± 2.23	0.563	1.52
8 weeks	111	432.4 ± 12.4 ^b	288	502.4 ± 6.95 ^a	0.0001	16.2
12 weeks	171	735 ± 14.0 ^b	283	810 ± 8.69 ^a	0.0001	10.2
16 weeks	99	906.2 ± 17.1 ^b	250	1008 ± 10.7 ^a	0.0001	10.1
20 weeks	202	1100 ± 15.2 ^b	314	1209 ± 9.91 ^a	0.0001	9.02
40 weeks	175	1486 ± 21.1 ^b	266	1598 ± 14.5 ^a	0.0001	7.01
Shank length (mm)						
8 weeks	84	52.82 ± 0.81 ^b	199	67.43 ± 0.46 ^a	0.0001	27.7
12 weeks	171	67.11 ± 0.53 ^b	283	83.68 ± 0.33 ^a	0.0001	24.7
20 weeks	202	70.03 ± 0.39 ^b	314	90.25 ± 0.25 ^a	0.0001	22.4
40 weeks	175	71.07 ± 0.37 ^b	266	90.69 ± 0.26 ^a	0.0001	21.6
ASM (days)	180	173.9 ± 1.44 ^b	287	168.7 ± 0.98 ^a	0.003	-3.08
SEP 40 weeks (N)	180	62.05 ± 1.83	287	65.75 ± 1.25	0.097	5.63
Egg wt. at 32 week (g)	160	45.57 ± 0.32 ^b	245	46.64 ± 0.22 ^a	0.006	2.23
Egg wt. at 40 week (g)	130	47.97 ± 0.36 ^b	202	49.27 ± 0.26 ^a	0.004	2.64

Values are expressed as means ± standard error.

^{a, b} Means with different superscripts within a row differ significantly; N, number of observations; ASM, age at sexual maturity; SEP, survivor egg production.

following standard procedures[22]. General linear model analysis (SPSS) was used to study the effects of dwarfism on the different traits. Sex-wise analyses were performed for body weight and shank length.

$$Y_{ijkl} = \mu + A_i + B_j + C_k + e_{ijkl}$$

where Y_{ijkl} = record for the l^{th} individual of the k^{th} (dwarf or normal) phenotype of the j^{th} hatch of the i^{th} generation; μ = population mean; A_i is the fixed effect of the i^{th} generation ($i=1$ to 2); B_j is the fixed effect of the j^{th} hatch ($j=1$ to 2); C_k is the fixed effect of the phenotype ($k=1$ (dwarf) or 2 (normal)); and e_{ijkl} is the random error, assumed to be normally and independently distributed with mean zero and constant variance.

Results

Growth traits

Growth traits, such as body weight and shank length, were studied sex-wise from day 0 to 40 weeks of age. Body weight on day 0 and on the 4th week was not affected by dwarfism in either male or female birds. However, from 8 weeks onward, body weight became significantly lower in dwarf birds than in their normal counterparts and persisted as such up to the age of 40 weeks (Tables 1 and 2). Similarly, shank lengths recorded at 8, 12, 20, and 40 weeks were significantly lower in both male and female dwarf birds than in normal birds (Tables 1 and 2).

Production traits

Age at sexual maturity (age at first egg) was significantly ($P < 0.003$) higher in dwarf hens than in normal hens; whereas egg

production up to 40 weeks of age was only slightly lower ($P < 0.097$) in dwarf hens. Egg weights recorded at 32 ($P < 0.006$) and 40 ($P < 0.004$) weeks were lower in dwarf hens than in non-dwarf hens (Table 2).

Egg quality traits

At the age of 40 weeks, dwarfism did not seem to affect internal egg quality traits. Instead, it affected external egg quality traits, with egg weight being significantly ($P < 0.005$) higher in normal hens and the shape index being significantly ($P < 0.038$) higher in dwarf hens. Shell thickness was significantly ($P < 0.014$) higher; while shell color was significantly darker ($P < 0.0001$) and red ($P < 0.0001$), with less of a hue ($P < 0.0001$) in dwarf hen eggs. The SCI ($P < 0.001$) and E ($P < 0.0001$) values were significantly lower in the eggs of dwarf hens than in normal hens (Table 3).

Egg weight correlated positively with lightness (L^*), hue, SCI, and E values, and negatively with redness (a^*) and yellowness (b^*) in the eggs of dwarf hens (Table 4). No such relationship was observed between egg weight and shell color characteristics in the eggs of normal hens (Table 5). All other shell color characteristics correlated similarly between dwarf and normal hens.

Reproduction traits

Fertility and hatchability (of the fertile egg set and total egg set) was significantly ($P < 0.0001$) lower in dwarf dams mated with dwarf sires than in normal sires mated with normal dams, dwarf sires mated with normal dams (cross), and normal sires mated with dwarf dams (reciprocal cross). There was no significant difference in hatchability and fertility among these last three

Table 3. Egg quality traits of dwarf and normal Nicobari birds at 40 weeks of age.

Traits	N	Dwarf	N	Normal	P value
External egg quality traits					
Egg weight (g)	158	48.24 ± 0.30 ^b	217	49.39 ± 0.27 ^a	0.005
Shape index	122	76.23 ± 0.37 ^a	142	75.20 ± 0.33 ^b	0.038
Shell thickness (mm)	73	0.409 ± 0.003 ^a	73	0.397 ± 0.003 ^b	0.014
Internal egg quality traits					
Yolk index	72	0.45 ± 0.004	69	0.46 ± 0.004	0.548
Yolk color	72	8.11 ± 0.20	69	7.87 ± 0.20	0.387
Albumen index	72	0.07 ± 0.03	73	0.08 ± 0.02	0.382
Haugh unit	72	77.69 ± 1.13	73	78.97 ± 1.00	0.398
Cal HU	72	80.11 ± 1.08	73	81.63 ± 0.93	0.284
Yolk weight (g)	72	15.81 ± 0.21	69	16.07 ± 0.19	0.358
Albumen weight (g)	72	27.52 ± 0.32	69	27.45 ± 0.37	0.891
Shell weight (g)	73	4.42 ± 0.06	73	4.40 ± 0.05	0.841
Yolk percentage	72	33.14 ± 0.38	69	33.64 ± 0.40	0.419
Albumen percentage	72	57.60 ± 0.37	69	57.17 ± 0.40	0.372
Shell percentage	72	9.26 ± 0.12	72	9.24 ± 0.10	0.436
Yolk-to-albumen ratio	72	0.58 ± 0.01	68	0.59 ± 0.01	0.454
Egg shell color characteristics					
L*	112	72.6 ± 0.43 ^b	171	74.8 ± 0.28 ^a	0.0001
a*	112	7.84 ± 0.28 ^a	171	6.78 ± 0.17 ^b	0.0001
b*	112	20.5 ± 0.33	171	20.0 ± 0.24	0.204
Hue	112	1.21 ± 0.01 ^b	171	1.25 ± 0.01 ^a	0.0001
Chroma	112	22.0 ± 0.39	171	21.1 ± 0.27	0.055
SCI	112	44.25 ± 0.95 ^b	171	48.07 ± 0.63 ^a	0.001
E value	112	76.09 ± 0.32 ^b	171	77.90 ± 0.22 ^a	0.0001

Values are expressed as means ± standard error.

^{a, b} Means with different superscripts within a row differ significantly; Cal HU: calculated Haugh unit; L*, lightness; a*, redness; b*, yellowness; SCI, shell color index; N, number of observations.

Table 4. Pearson correlation coefficients of egg shell color parameters in dwarf Nicobari hens (N=85).

	Egg wt.	L*	a*	b*	Hue	Chroma	SCI	E value
Egg wt.	1							
L*	0.40**	1						
a*	-0.44**	-0.85**	1					
b*	-0.37**	-0.82**	0.81**	1				
Hue	0.33**	0.50**	-0.78**	-0.33**	1			
Chroma	-0.42**	-0.83**	0.92**	0.96**	-0.57**	1		
SCI	0.43**	0.95**	-0.95**	-0.92**	0.61**	-0.96**	1	
E value	0.32**	0.96**	-0.71	-0.70	0.34	-0.66	0.83	1

L*, lightness; a*, redness; b*, yellowness; SCI, shell color index. **, $P < 0.01$.

types of mating (Table 6).

Semen quality traits

No significant differences in semen quality were observed between dwarf and normal cocks; only the live sperm count was significantly ($P < 0.016$) lower in the former (Table 7).

Genetic determination of inheritance

The dwarf gene in the Nicobari breed appears to follow an au-

tosomal recessive inheritance. However, sequencing of a 264-bp fragment revealed no mutation (TGG to TAG) in the TMEM263 (*adw*) gene of dwarf birds, suggesting that the dwarf character was not due to the *adw* mutation. To rule out the possibility of sex-linked inheritance, we amplified a 253-bp fragment targeting a deletion in the growth hormone receptor gene specific for the sex-linked dwarf (*dw*) allele[21], but failed to obtain a corre-

Table 5. Pearson correlation coefficients of egg shell color parameters in normal Nicobari hens (N=144).

	Egg wt.	L*	a*	b*	Hue	Chroma	SCI	E value
Egg wt.	1							
L*	0.15	1						
a*	-0.07	-0.85**	1					
b*	-0.14	-0.75**	0.78**	1				
Hue	0.02	0.70**	-0.89**	-0.43**	1			
Chroma	-0.13	-0.80**	0.85**	0.99**	-0.54**	1		
SCI	0.14	0.94**	-0.93**	-0.91**	0.70	-0.94**	1	
E value	0.15	0.98**	-0.77**	-0.61**	0.68**	-0.67**	0.87**	1

L*, lightness; a*, redness; b*, yellowness; SCI, shell color index. **, $P < 0.01$.

Table 6. Fertility and hatchability traits of different crosses.

Traits	N	Dwarf sire × Dwarf dam	N	Normal sire × Normal dam	N	Dwarf sire × Normal dam	N	Normal sire × Dwarf dam	P value
Fertility (%)	109	80.06±1.89 ^b	119	92.92±1.25 ^a	155	93.90±1.02 ^a	42	94.91±1.69 ^a	0.0001
Hatchability (%)									
FES	109	87.62±1.82 ^b	119	97.14±0.83 ^a	155	95.88±0.94 ^a	42	93.49±2.15 ^a	0.0001
TES	109	70.07±2.21 ^b	119	90.11±1.40 ^a	155	90.00±1.34 ^a	42	88.82±2.69 ^a	0.0001

Values are expressed as means ± standard error.

FES, fertile egg set; TES, total egg set; N, number of dams (observations).

^{a, b} Means bearing different superscripts within a row differ significantly.

Table 7. Semen quality of dwarf and normal Nicobari birds.

Traits	Dwarf (N=39)	Normal (N=40)	P value
Semen volume (mL)	0.34 ± 0.02	0.38 ± 0.02	0.119
Motility (%)	60.34 ± 0.96	61.01 ± 0.77	0.551
Sperm concentration (billion/mL)	4.86 ± 0.22	4.65 ± 0.22	0.816
MTT (nm formazan/min/million sperm)	23.98 ± 1.08	24.61 ± 1.06	0.679
Live sperm (%)	68.08 ± 0.82 ^b	71.32 ± 1.04 ^a	0.016
Abnormal sperm (%)	7.75 ± 0.44	7.17 ± 0.44	0.364

Values are expressed as means ± standard error.

3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, MTT

^{a, b} Means with different superscripts within a row differ significantly.

sponding product in dwarf birds. Hence, segregation of dwarfism in the Nicobari breed appears to follow an unprecedented pattern.

Discussion

The present study assessed the effect of dwarfism on growth, production, reproduction, and egg quality in the Nicobari chicken breed. A significant drop in body weight was observed in dwarf birds from week 8 onwards. The reduction was higher in female birds (9% to 16%) than in male birds (5% to 9%) between 8 and 20 weeks of age. An even larger drop was observed for shank length, was similar (22%–27%) in male and female birds. Thus, dwarfism had a low to moderate impact on body weight, but a greater one on shank length. Previous studies reported that the sex-linked dwarf gene (*dw*) reduced live body weight by 31%–33% in adult hens[6] and 37.4% in White Plymouth Rock fe-

males[23]. Chen et al.[21] reported a 25.3% to 37.4% reduction in body weight at 8 and 20 weeks of age; whereas shank length was decreased by only 14.2% and 21.7% in a Taiwanese strain carrying the *dw* gene as compared to normal Taiwanese country birds. Furthermore, body weight declined was apparent at week 2 (13.4%) itself in Taiwanese chickens. Similarly, Mincheva et al.[24] reported a significant loss of body weight (29.15%) and shank length (20.17%) owing to the *dw* gene in hens. The autosomal dwarf gene (*adw*) was also reported to reduce body weight by approximately 30% in both males and females of K-strain of White Leghorns[9]. Hence, dwarf genes (both *dw* and *adw*) exerted a greater effect on body weight (25% to 37%) than on shank length (20%).

In sex-linked dwarfism, the effect on body weight is evident as early as at 2 weeks of age[21]; whereas autosomal dwarfism

(*adw*) starts affecting body weight from the embryonic stage[9]. Interestingly, the dwarf character in the Nicobari breed had an entirely different effect on body weight and shank length, and became evident only at the age of 8 weeks. These findings suggest that the dwarf character of the Nicobari breed differs from that of sex-linked (*dw*) and autosomal (*adw*) mutations.

A significant delay in the age at the first egg was observed. This was followed by a slight, albeit non-significant, decline in the number of eggs produced over 40 weeks and a significant reduction in egg weight at 32 and 40 weeks of dwarf hens compared to normal hens. Similar findings were reported earlier, whereby a delay of 1 to 2 weeks for the first egg was detected in sex-linked dwarf hens, although it was not accompanied by decreased egg production[6]. Mincheva et al.[24] also reported a significant delay in age at first egg and a lower egg weight (5.72%), but no effect on the egg-laying rate, in dwarf hens compared to normal ones. Autosomal dwarfism has been reported to delay the age at sexual maturity by 2 weeks[9]. The delayed age at first egg observed in dwarf hens could be due to lower body weight and, hence, delayed maturity, rather than a direct effect of the dwarf gene on reproductive organs[9].

Dwarf characteristics had no effect on internal egg quality traits, although the eggs of dwarf hens had a significantly higher shape index and shell thickness. Sex-linked dwarfism has been reported to have no effect on internal egg quality traits[24]. Interestingly, the eggshells of dwarf hens were significantly darker and red, with less hue than those of normal hens. Significantly lower SCI and E values indicated that the eggshells of dwarf hens were darker than those of normal hens of the Nicobari breed. Additionally, the smaller size of dwarf hens produce smaller eggs, in which pigment may become denser. Several other factors affect eggshell color, including breed, strain, age, diet, and environment.

Egg weight was significantly positively correlated with eggshell color characteristics (L^* , hue, SCI, and E value) and negatively correlated with a^* , b^* , and chroma in eggs of dwarf hens. However, no such correlation was observed in eggs of normal hens. Similarly, Goger et al.[25] observed a positive correlation between eggshell color and L^* (lightness). Why the relationship between eggshell color and egg weight differs between dwarf and normal hens remains to be determined, but might be related to disparities in egg size.

A significant drop in fertility (12.86 basis points (bps)/16.06%) and hatchability (9.52 bps/10.86% and 20.04 bps/28.6% for the fertile and total egg sets) was observed in dwarf dams mated with dwarf sires as compared to matings between normal sires $\times$ normal dams, normal dams $\times$ dwarf sires (cross), and dwarf dams $\times$ normal sires (reciprocal cross). This finding suggests that male and female determinants synergistically affect the fertility and hatchability of dwarf birds. It has been hypothesized that the factors/genes responsible for fertility and hatchability in dwarf birds are recessive in nature; hence, their effects are expressed only under homozygous (dwarf $\times$ dwarf) conditions. The genes responsible for dwarfism affect the reproductive traits of Nico-

bari dwarf-type birds either directly or indirectly. These observations are contrary to previous reports of similar fertility and hatchability in sex-linked dwarf and normal dams[6,26]. One study reported improved hatchability (total egg set) in sex-linked dwarf dams compared with normal dams[24]. The lack of a significant difference in fertility and hatchability between dwarf sire $\times$ normal dam, normal sire $\times$ dwarf dam and normal sire $\times$ normal dam indicates that either male or female factor alone may not be contributing to reduced fertility. This finding is further supported by the absence of any detectable difference in semen quality between dwarf and normal sires.

Inheritance of dwarfism in the Nicobari breed appears to follow a recessive pattern. To exclude the possibility of sex-linked inheritance, dwarf sires were mated with normal dams. This cross must produce male chicks of the normal phenotype and female chicks of the dwarf phenotype to confirm sex-linked inheritance. Our results did not confirm sex-linked inheritance. In this breed, it was difficult to accurately identify all dwarf individuals (sex-wise) until birds reached at least 14–16 weeks of age. During this period, mortality due to nonspecific causes was recorded, and normal-looking male birds had to be disposed of because of space constraints. Therefore, it was not possible to determine the exact number of dwarf offspring obtained from each cross in first-generation breeding experiments. Instead, in the cross between normal males and dwarf females, we obtained 68% dwarf and 32% normal male offspring, plus 64% dwarf and 36% normal female offspring. Furthermore, in sex-linked inheritance, normal $\times$ normal crosses are expected to produce all progenies (both males and females) of the normal phenotype. However, we still recorded 9% (male) to 10% (female) dwarf offspring in this cross because of recessive alleles. As expected, the dwarf $\times$ dwarf crosses produced male and female dwarf progenies. Therefore, we ruled out the possibility of sex-linked inheritance. After collecting fertility and hatchability data, the breeding experiment in the second generation was discontinued owing to heavy mortality.

Finally, the possibility of sex-linked inheritance of dwarfism (*dw*) in the Nicobari breed was excluded by molecular analysis. Nucleotide sequencing of part of the TMEM263 gene also ruled out the possibility of autosomal dwarfism (*adw*) caused by a mutation in the TMEM263 gene. Therefore, autosomal recessive dwarfism observed in this breed appears to arise from a novel mechanism. This is further supported by the fact that the observed effects on growth and reproductive traits differ from those of dwarfism caused by *dw* and *adw* mutations.

In conclusion, the present study indicates that naturally segregating dwarfism in the Nicobari breed is unique and appears to be recessive and autosomal. It has a significant influence on growth, production, reproduction, and external egg quality traits, but no effect on internal egg or semen quality. Further studies are required to determine the molecular basis of dwarfism and its impact on physiological traits, heat stress resilience, feed intake, feed conversion ratio, and carcass and meat quality characteristics in the Nicobari chicken breed.

Acknowledgments

The authors gratefully acknowledge the technical help of the hatchery and farm personnel in hatching chicks and caring for the birds. The experiment was conducted as part of an in-house project (ANSCDPRSIL201500300052) at the institute. Authors express gratitude to Dr. Wu Zhou, Roslin institute for kind correspondence.

Author Contributions

Santosh Haunshi: conceptualized the study, designed and conducted the experiment, analysed the data and wrote the original draft. Shanmugam Murugesan: studied the semen quality parameters, analysed and interpreted the results. Leslie L.L. Prince: analysed and curated the data; Rajkumar Ullengala: supervised the experiment and reviewed and edited the manuscript. Rudra Nath Chatterjee: secured the funding, arranged the resources and responsible for the project administration. All authors have read, reviewed and agreed to the published version of the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] Haunshi S, Sharma D, Nayal LMS, Singh DP and Singh RV. Effect of naked neck gene (NA) and frizzle gene (F) on immunocompetence in chickens. *Br Poult Sci*, **43**: 28–32. 2002. <https://doi.org/10.1080/00071660120109863>, PMID:12003334
- [2] Arora G, Mishra SK, Nautiyal B, Pratap SO, Gupta A, Beura CK and Singh DP. Genetics of hyperpigmentation associated with the Fibromelanosis gene (*Fm*) and analysis of growth and meat quality traits in crosses of native Indian Kadaknath chickens and non-indigenous breeds. *Br Poult Sci*, **52**: 675–685. 2011. <https://doi.org/10.1080/00071668.2011.635637>, PMID:22221233
- [3] Shinde SS, Sharma A and Vijay N. Decoding the fibromelanosis locus complex chromosomal rearrangement of black-bone chicken: genetic differentiation, selective sweeps and protein-coding changes in Kadaknath chicken. *Front Genet*, **14**: 1180658. 2023. <https://doi.org/10.3389/fgene.2023.1180658>, PMID:37424723
- [4] Wu Z, Bortoluzzi C, Derks MFL, Liu L, Bosse M, Hiemstra SJ, Groenen MAM and Crooijmans RPMA. Heterogeneity of a dwarf phenotype in Dutch traditional chicken breeds revealed by genomic analyses. *Evol Appl*, **14**: 1095–1108. 2021. <https://doi.org/10.1111/eva.13183>, PMID:33897823
- [5] Jin S, Zhu F, Wang Y, Yi G, Li J, Lian L, Zheng J, Xu G, Jiao R, Gong Y, Hou Z and Yang N. Deletion of Indian hedgehog gene causes dominant semi-lethal Creeper trait in chicken. *Sci Rep*, **6**: 30172. 2016. <https://doi.org/10.1038/srep30172>, PMID:27439785
- [6] Merat P. The sex-linked dwarf gene in the broiler chicken industry. *Worlds Poult Sci J*, **40**: 10–18. 1984. <https://doi.org/10.1079/WPS19840002>
- [7] Tixier-Boichard M. From phenotype to genotype: major genes in chickens. *Worlds Poult Sci J*, **58**: 65–75. 2002. <https://doi.org/10.1079/WPS20020008>
- [8] Ouyang J, Xie L, Nie Q, Zeng H, Peng Z, Zhang D and Zhang X. The effects of different sex-linked dwarf variations on Chinese native chickens. *J Integr Agric*, **11**: 1500–1508. 2012. [https://doi.org/10.1016/S2095-3119\(12\)60150-6](https://doi.org/10.1016/S2095-3119(12)60150-6)
- [9] Cole RK. An autosomal dwarfism in the domestic fowl. *Poult Sci*, **79**: 1507–1516. 2000. <https://doi.org/10.1093/ps/79.11.1507>, PMID:11092317
- [10] Wu Z, Derks MFL, Dibbits B, Megens HJ, Groenen MAM and Crooijmans RPMA. A novel loss-of-function variant in transmembrane protein 263 (TMEM263) of autosomal dwarfism in chicken. *Front Genet*, **9**: 193. 2018. <https://doi.org/10.3389/fgene.2018.00193>, PMID:29930570
- [11] Chatterjee RN and Yadav SP. Farming system of Nicobari fowl – an endangered breed of Andaman and Nicobar Islands, India. *Worlds Poult Sci J*, **64**: 245–256. 2008. <https://doi.org/10.1017/S0043933908000032>
- [12] Kundu A, De AK, Kundu MS, Sunder J, Jeyakumr S and Sujatha T. Production performance of indigenous Nicobari fowls, Vanaraja and their various F1 crosses under hot humid climate of Andaman and Nicobar Islands, India. *Indian J Anim Sci*, **85**: 172–177. 2015. <https://doi.org/10.56093/ijans.v85i2.46611>
- [13] Haunshi S, Burramsetty AK, Kannaki TR and Rajkumar U. Survivability, immunity, growth and production traits in indigenous and White Leghorn breeds of chicken. *Br Poult Sci*, **60**: 683–690. 2019. <https://doi.org/10.1080/00071668.2019.1639139>, PMID:31280601
- [14] Burrows WH and Quinn JP. The collection of spermatozoa from the domestic fowl and turkey. *Poult Sci*, **16**: 19–24. 1937. <https://doi.org/10.3382/ps.0160019>
- [15] Chaudhuri D and Lake PE. A new diluent and methods of holding semen for up to 17 hours at high temperature. *Proc. 28th World's Poultry Congress*. Nagoya, Japan. pp. 591–593, 1988.
- [16] Taneja GC and Gowe RS. Spermatozoa concentration in the semen of two breeds of fowl estimated by three different methods. *Poult Sci*, **40**: 608–615. 1961. <https://doi.org/10.3382/ps.0400608>
- [17] Hazary RC, Chaudhuri D and Wishart GJ. Application of an MTT reduction assay for assessing sperm quality and predicting fertilising ability of domestic fowl semen. *Br Poult Sci*, **42**: 115–117. 2001. <https://doi.org/10.1080/713655009>, PMID:11337959
- [18] Campbell RC, Hancock JL and Rothschild L. Counting live and dead bull spermatozoa. *J Exp Biol*, **30**: 44–49. 1953. <https://doi.org/10.1242/jeb.30.1.44>
- [19] Cavero D, Schmutz M, Icken W and Preisinger R. Attractive eggshell color as a breeding goal. *Lohmann Inform*, **47**: 15–21. 2012.
- [20] Baylan M, Celik LB, Akpinar GC, Alasahan S, Kucukgul A and Dogan SC. Influence of eggshell colour on egg yolk antibody level, incubation results, and growth in broiler breeders. *Rev Bras Zootec*, **46**: 105–112. 2017. <https://doi.org/10.1590/s1806-92902017000200004>
- [21] Chen CF, Chen YH, Tixier-Boichard M, Cheng PY, Chang CS, Tang PC and Lee YP. Effects of the Chicken Sex-linked Dwarf Gene on rowth and Muscle Development. *Asian-Australas J Anim Sci*, **22**: 937–942. 2009. <https://doi.org/10.5713/ajas.2009.80689>

- [22] Snedecor GW and Cochran WG. Statistical Methods, 8th Edition Wiley-Blackwell, ISBN: 978-0-813-81561-9, USA, 1989.
- [23] Knížetová H. Effects of the sex-linked dwarf gene (dw) on skeletal muscle cellularity in broiler chickens. *Br Poult Sci*, **34**: 479–485. 1993. <https://doi.org/10.1080/00071669308417603>, PMID:8358635
- [24] Mincheva N, Oblakova M, Hristakieva P, Ivanova I and Lalev M. Effect of sex-linked dwarf gene on exterior appearance, productive performance and egg characteristics in a colored broiler dam line. *Biotechnol Anim Husb*, **31**: 163–174. 2015. <https://doi.org/10.2298/BAH1502163M>
- [25] Goger H, Demirtas SE and Yurtogullari S. A selection study for improving eggshell colour in two parent lines of laying hens and their hybrids. *Ital J Anim Sci*, **15**: 390–395. 2016. <https://doi.org/10.1080/1828051X.2016.1215232>
- [26] Tahir M, Cervantes H, Farmer CW, Shim MY and Pesti GM. Broiler performance, hatching egg, and age relationships of progeny from standard and dwarf broiler dams. *Poult Sci*, **90**: 1364–1370. 2011. <https://doi.org/10.3382/ps.2010-01165>, PMID:21597080